Harmonic Form

Exam code: 9MA0

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Harmonic form

What is harmonic form?

- If ***a*** and ***b*** are positive constants, then
  - $a \sin x + b\cos x$ can be rewritten as $R \sin (x + \alpha)$
  - $a \sin x - b\cos x$ can be rewritten as $R \sin (x - \alpha)$
  - $a \cos x + b\sin x$ can be rewritten as $R \cos (x - \alpha)$
  - $a \cos x - b\sin x$ can be rewritten as $R \cos (x + \alpha)$
- The forms with R and α are called **harmonic forms**
  - It is the reverse of expanding the right-hand side out using compound angle formulae
  - It can be seen as factorising
- In all four cases $R > 0$ and $0 < \alpha < 90°$, with:

$$a = R\cos\alpha$$
$$b = R\sin\alpha$$
$$R = \sqrt{a^2 + b^2}$$

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- To find α, first divide the **b** expression by the **a** expression
  - The R's cancel
  - Use the identity $\sin\alpha / \cos\alpha = \tan\alpha$
  - You get $b / a = \tan\alpha$
  - Then use $\tan^{-1}(b/a)$ to find α

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- To rewrite the expression use the appropriate **compound angle formula** and equate coefficients

e.g. $5\cos x - 12\sin x$

THIS IS CASE 4 ABOVE, WITH $a=5$ AND $b=12$

$\cos(x+\alpha) = \cos\alpha\cos x - \sin\alpha\sin x$

EXPAND USING COMPOUND ANGLE FORMULA FOR cos

SO $R\cos(x+\alpha) = R\cos\alpha\cos x - R\sin\alpha\sin x$

COMPARE TO ORIGINAL EXPRESSION:

$R\cos\alpha = 5 \qquad R\sin\alpha = 12$

$R\cos\alpha = a$
$R\sin\alpha = b$

$\frac{R\sin\alpha}{R\cos\alpha} = \tan\alpha = \frac{12}{5}$

$\alpha = \tan^{-1}\left(\frac{12}{5}\right) = 67.38^\circ$

ALSO $(R\cos\alpha)^2 + (R\sin\alpha)^2 = 5^2 + 12^2$

$R^2\cos^2\alpha + R^2\sin^2\alpha = 25 + 144$

$R^2(\cos^2\alpha + \sin^2\alpha) = 169$

$\cos^2\alpha + \sin^2\alpha = 1$

$R^2 = 169$

$R = 13$

$R = \sqrt{a^2 + b^2}$

$5\cos x - 12\sin x = 13\cos(x + 67.38^\circ)$

$a\cos x - b\sin x = R\cos(x+\alpha)$

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Examiner Tips and Tricks

- There is more than one way to rewrite a given expression, and the question in the exam will tell you which form to use.

Worked Example

where $R > 0$ and $0 < \alpha < 90°$.

b) Hence solve $\sin\theta - 4\cos\theta = 3$ for $0 \le \theta < 360°$.

COMPOUND ANGLE FORMULA FOR sin

a) $R\sin(\theta - \alpha) = R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$

$R\cos\alpha = 1 \qquad R\sin\alpha = 4$

$R\cos\alpha = a$
$R\sin\alpha = b$

$\frac{R\sin\alpha}{R\cos\alpha} = \tan\alpha = 4$

$\alpha = \tan^{-1}(4) = 75.96°$

$R = \sqrt{1^2 + 4^2} = \sqrt{17}$

$R = \sqrt{a^2 + b^2}$

$\sin\theta - 4\cos\theta = \sqrt{17}\sin(\theta - 75.96°)$

b) $\sqrt{17}\sin(\theta - 75.96°) = 3$

$\sin(\theta - 75.96°) = \frac{3}{\sqrt{17}}$

$(\theta - 75.96°) = 46.69°$

OR $(\theta - 75.96°) = 180 - 46.69 = 133.31°$

USE $\sin^{-1}$ ON YOUR CALCULATOR TO FIND THE PRINCIPAL VALUE OF θ

THE SOLUTIONS ARE
$\theta = 122.65°$ OR $209.27°$

THEN FIND OTHER SOLUTIONS IN THE REQUIRED RANGE

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